



School of Computer Science and Engineering

Winter Semester 2022-2023

Continuous Assessment Test – 1 Answer Key

SLOT C2 +TC2

Programme Name & Branch : B.TECH CSE

Course Name & code: BCSE202L Data Structures and Algorithms

Class Number (s) : VL2022230505827, VL2022230505830, VL2022230505833, VL2022230505835, VL2022230505854, VL2022230505863, VL2022230505954, VL2022230506219, VL2022230506332, VL2022230506588, VL2022230506597, VL2022230507613

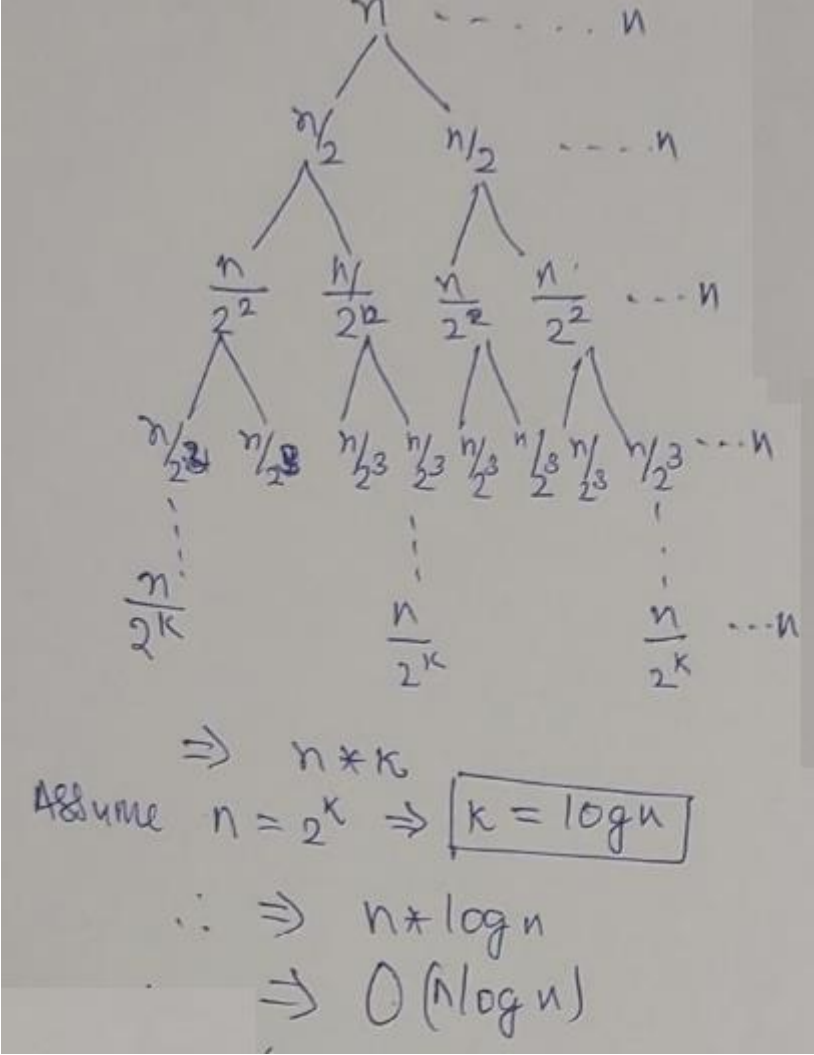
Faculty Name (s) : ALL

Exam Duration: 90 Min.

Maximum Marks: 50

Answer ALL Questions

Q.No.	Question	Max Marks	CO	BL
1.	a) Compute the Time complexity of the following function <pre>Function() { int i,j,k; for(i=n;i>=n/2;i--) for(j=1;j<=n;j=3*j) for(k=1;k<=n;k=k*3) printf("Hello"); }</pre> Solution : $n/2 * \log_3 n * \log_3 n = n/2 (\log_3 n)^2$ $= O(n (\log_3 n)^2)$ b) Solve the following Recurrence Relation using Recursive Tree Method $T(n) = \begin{cases} 1 & n=1 \\ 2T(n/2) + n & n>1 \end{cases}$	3+7	CO1	BL4

	 $\Rightarrow n * k$ Assume $n = 2^k \Rightarrow k = \log n$ $\Rightarrow n * \log n$ $\Rightarrow O(n \log n)$			
2.	Let $T(n) = 9T(n/3) + n^2 \log n$ be the given recurrence relation. Solve the given Recurrence Relation using a) Backward Substitution Method Solution Apply stepwise and compute $\Theta(n^2 \log^2 n)$ b) Masters Theorem Solution : $A=9$ $b=3$ $k=3$ $p=1$ $A=b^k$ $9=3^3$ Case 2 a) holds if $a = b^k$, then (a) if $p > -1$, then $T(n) = \Theta(n^{\log_b a} \log^{p+1} n)$ $= \Theta(n^{\log_3 9} \log^{1+1} n) = \Theta(n^2 \log^2 n)$ $= \Theta(n^2 \log^2 n)$	7+3	CO1	BL3

3.	Illustrate how the given infix expression is converted into a Postfix expression. Make sure to show all the steps involved in the process. (A – B) / (((D + E) * F) – G) Solution: Postfix conversion: <table> <tr> <th><u>ch</u></th> <th><u>action</u></th> <th><u>operator stack</u></th> <th><u>Postfix</u></th> </tr> <tr> <td>(</td> <td>push</td> <td>(</td> <td></td> </tr> <tr> <td>a</td> <td>output</td> <td>(</td> <td>a</td> </tr> <tr> <td>–</td> <td>push</td> <td>(–</td> <td>a</td> </tr> <tr> <td>b</td> <td>output</td> <td>(–</td> <td>a b</td> </tr> <tr> <td>)</td> <td>pop until (</td> <td></td> <td>a b –</td> </tr> <tr> <td>/</td> <td>push</td> <td>/</td> <td>a b –</td> </tr> </table> Likewise all the other steps should be carried out to get the final postfix expression as a b – d e + f * g - /	<u>ch</u>	<u>action</u>	<u>operator stack</u>	<u>Postfix</u>	(	push	(		a	output	(	a	–	push	(–	a	b	output	(–	a b	)	pop until (		a b –	/	push	/	a b –	10	CO2	BL5
<u>ch</u>	<u>action</u>	<u>operator stack</u>	<u>Postfix</u>																													
(	push	(																														
a	output	(	a																													
–	push	(–	a																													
b	output	(–	a b																													
)	pop until (		a b –																													
/	push	/	a b –																													
4.	Narrate the following operations in a circular queue of size 10. Show the status of circular queue for each of the print operation <pre>for (int k = 9; k >=0; k--) printf(enqueue(k)); for (int k = 0; k < 4; k++) { printf(dequeue()); printf(enqueue(dequeue())); printf(dequeue()); }</pre> Solution /* enqueueer */ F R 9 8 7 6 5 4 3 2 1 0 When K=0 F R 8 7 6 5 4 3 2 1 0 7 6 5 4 3 2 1 0 8 6 5 4 3 2 1 0 8 When K=1 F R 5 4 3 2 1 0 8 4 3 2 1 0 8 5 3 2 1 0 8 5 When K=2 F R 2 1 0 8 5 1 0 8 5 2 0 8 5 2	10	CO2	BL3																												

	When K=3 F R 8 5 2 5 2 8 2 8			
5.	a) Illustrate and Explain the working principle of Bubble Sort Algorithms using the following 7, 1, 3, 8, 2, 5 Numbers. Iteration -1 7 1 3 8 2 5 given input 1 7 3 8 2 5 1 3 7 8 2 5 1 3 7 8 2 5 1 3 7 2 8 5 1 3 7 2 5 8 Iteration -2 1 3 7 2 5 8 1 3 7 2 5 8 1 3 2 7 5 8 1 3 2 5 7 8 Iteration -3 1 3 2 5 7 8 1 2 3 5 7 8 1 2 3 5 7 8 Iteration -4 1 2 3 5 7 8 Since no swapping is incurred shows that the given elements are arranged in ascending order. Hence the process is terminated b) Estimate the total number of comparisons needed when N=7 to arrange the given elements in ascending order. (N is the total number of elements present in the given array) Solution : Number of elements present in the given array is 7 Number of comparisons involved in iteration 1 is 6 Number of comparisons involved in iteration 2 is 5 Number of comparisons involved in iteration 3 is 4 Number of comparisons involved in iteration 4 is 3 Number of comparisons involved in iteration 5 is 2 Number of comparisons involved in iteration 6 is 1 Total number of comparisons is 6+5+4+3+2+1 =21	7+3	CO3	BL3